

Industrial Internship Report on "Python Quiz Game Application"

Prepared by
[Shrutika Bhimashankar Patane]

Executive Summary

This report provides details of the Industrial Internship provided by Upskill Campus and The IoT Academy in collaboration with the industrial partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We were required to complete the project along with the report within a duration of 4 weeks.

My project was "**Python Quiz Game Application**", which is a Python-based application that automatically organizes files in a selected folder into categories such as images, documents, videos, and others. The system helps in reducing manual effort and keeps files properly arranged using a simple graphical interface.

This internship gave me a very good opportunity to understand real-world problems and develop practical solutions. It helped me improve my technical skills in Python, especially in file handling and GUI development. Overall, it was a very useful and learning experience for me.

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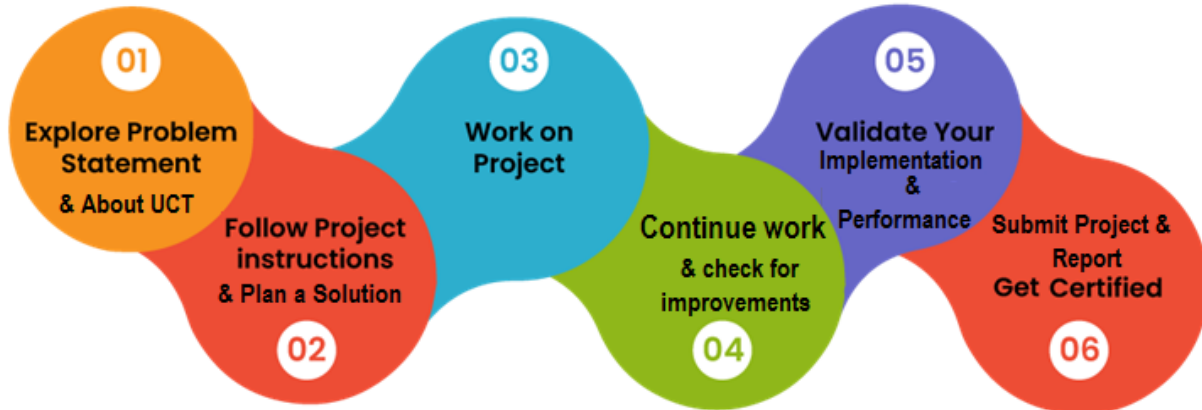
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1. Preface

This report summarizes the work completed during my four-week internship conducted through Upskill Campus in collaboration with The IoT Academy and UniConverge Technologies Pvt Ltd (UCT).

During this internship, I worked on a project titled “**Quiz Game.**” The objective of the project was to develop a Python-based quiz application that allows users to answer multiple-choice questions, calculate scores, and view their performance through a simple graphical interface.

This internship helped me improve my Python programming, GUI development, problem-solving, and debugging skills while providing valuable hands-on experience in real-world project development.



This internship provided me with a valuable opportunity to understand real-world software development practices and build a complete application from planning and design to implementation and testing. Through this project, I enhanced my knowledge of Python programming, GUI development, file handling, debugging, and problem-solving techniques.

Overall, this internship was a highly enriching learning experience that strengthened my technical skills, improved my confidence in software development, and prepared me for future industry-level projects.

1 Introduction

1.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



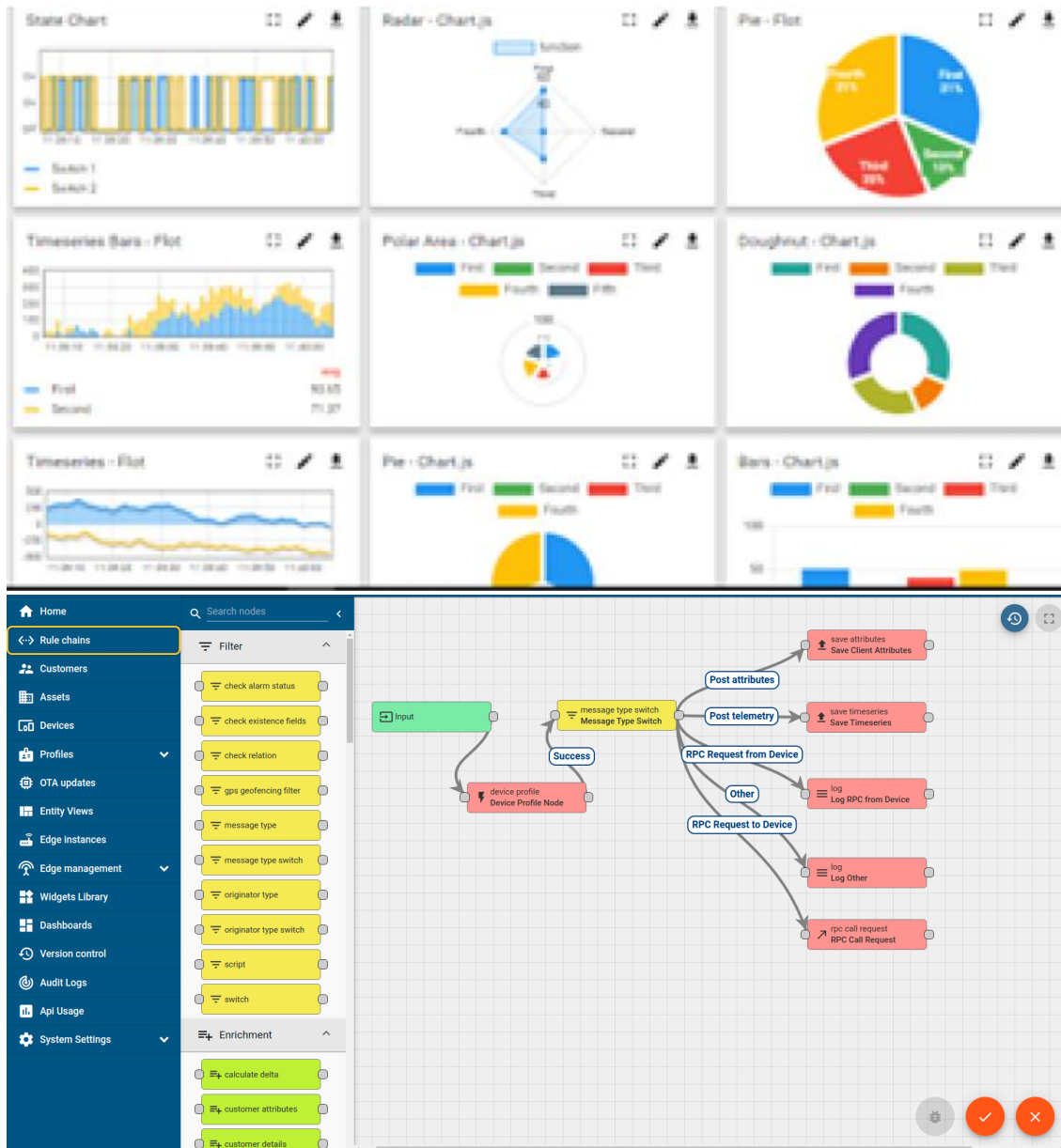
i. UCT IoT Platform (**uct Insight**)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



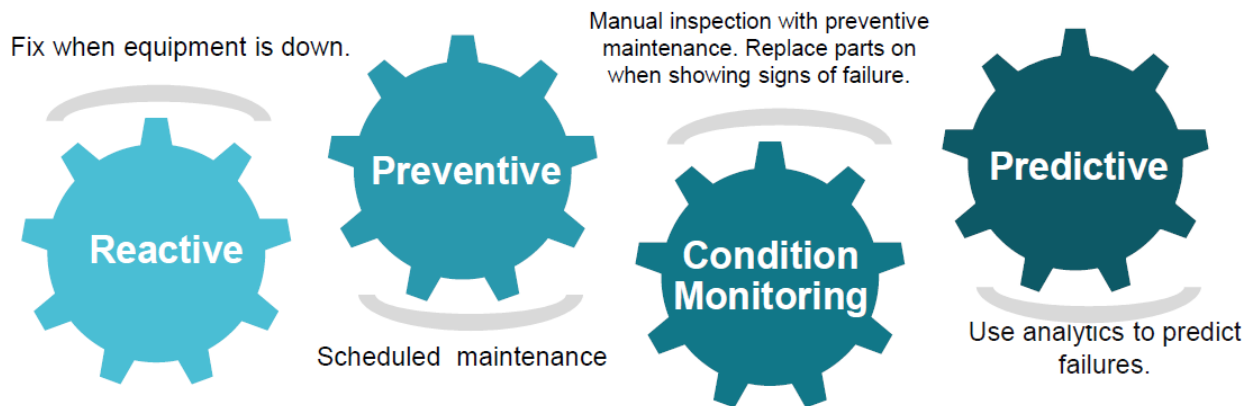


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

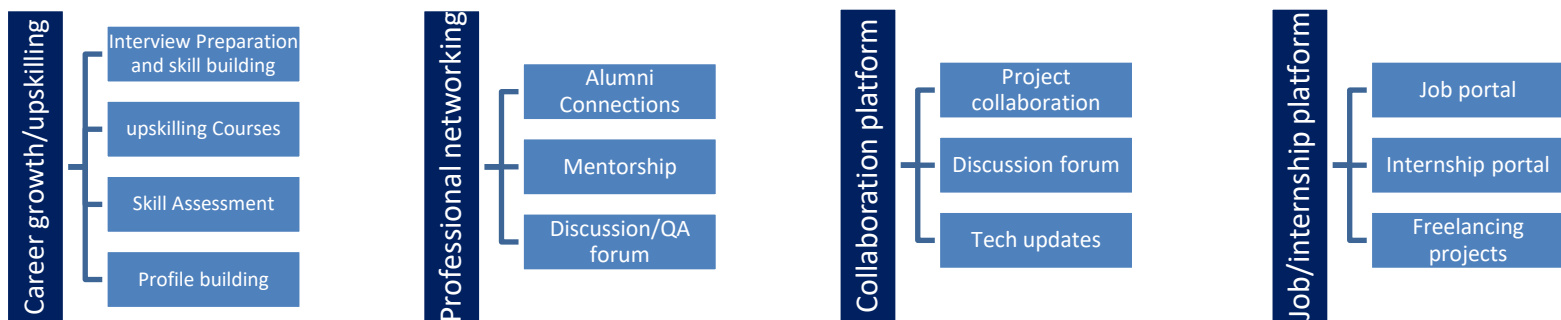
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



1.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



1.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

1.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

1.5 Reference

- [1] Python Official Documentation – <https://docs.python.org/3/>
[2] Tkinter Documentation – <https://docs.python.org/3/library/tkinter.html>
[3] Online tutorials and learning resources used during project development

1.6 Glossary

Terms	Acronym
Graphical User Interface	GUI
Operating System Module	OS
File Handling Library	Shutil
Integrated Development Environment	IDE
Portable Document Format	PDF

2 Problem Statement

The main objective of this project was to develop an interactive **Quiz Game** using Python.

Many traditional quiz systems are either manual or lack an engaging user interface, making it difficult for users to assess their knowledge effectively. There is a need for a simple and user-friendly application that can present questions, evaluate answers, and provide instant results.

To solve this problem, a Python-based Quiz Game was developed that displays multiple-choice questions, validates user responses, calculates scores, and shows the final performance. The application uses a graphical user interface (GUI) to provide a better user experience and make the quiz more interactive.

The solution is designed to be simple, efficient, and easy to use, allowing users to test their knowledge and receive immediate feedback on their performance.

3. Existing and Proposed Solution

3.1 Existing Solutions

Traditional quiz systems are often conducted manually using paper-based tests or simple applications with limited functionality. Some online quiz platforms exist, but they may require internet access and can be complex for beginners to use.

3.1.1 Limitations of Existing Solutions

- Limited user interaction and engagement
- Manual or less efficient score calculation
- Lack of interactive graphical interface
- Limited performance tracking features
- Dependence on internet connectivity in some systems

3.2 Proposed Solution

To overcome these limitations, a **Quiz Game** was developed using Python.

The application presents multiple-choice questions to users, validates their answers, calculates scores automatically, and displays the final result. A Graphical User Interface (GUI) developed using Tkinter makes the application easy to use and interactive.

The system also includes features such as random question selection, score calculation, percentage display, and timer-based quiz functionality to improve user experience.

3.3 Value Addition

The proposed solution provides several improvements over existing methods:

- Automatic score calculation
- User-friendly graphical interface
- Randomized question selection
- Timer-based quiz functionality
- Instant result and percentage display
- Improved user engagement and learning experience
- Faster and more accurate performance evaluation

2.1 Code submission (Github link)

2.2 Report submission (Github link) : first make placeholder, copy the link.

The **Quiz Game System** is designed to provide an interactive platform where users can answer multiple-choice questions and evaluate their knowledge. The system follows a simple flow starting from user login, presenting quiz questions, processing user responses, and generating the final result.

The design of the system is divided into three main stages:

- **Input Stage:** User enters details and starts the quiz.
- **Processing Stage:** System displays questions, validates answers, calculates scores, and manages the timer.
- **Output Stage:** Final score, percentage, and performance results are displayed to the user.

This step-by-step flow ensures that the system operates efficiently and provides an engaging and user-friendly quiz experience.

2.4 4.1 High Level Diagram

User → Login Screen → Load Questions → Display Questions → Answer Validation → Score Calculation → Result Display → Leaderboard

👉 Or in box format:

[User]

↓

[Login Screen]

↓

[Load Questions]

↓

[Display Questions]

↓

[Validate Answer]

↓

[Calculate Score]

↓

[Display Result]

↓

[Leaderboard / Score History]

Figure 1: High Level Diagram of the Quiz Game System

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2.5 4.2 Low Level Diagram

👉 Draw like this:

Start



Enter Username



Load Quiz Questions



Display Question



User Selects Answer



Validate Answer



Update Score



Next Question Available?



Yes → Display Next Question



No



Calculate Percentage



Display Final Result



Store Score / Update Leaderboard



End

👉 Below diagram, write:

Figure 2: Low Level Diagram of the Quiz Game System

2.6 5. Performance Test

The performance testing of the **Quiz Game System** was carried out to ensure that the application functions correctly and provides a smooth user experience. Testing was performed under different conditions to verify the accuracy of score calculation, question handling, timer functionality, and overall system responsiveness.

In real-world usage, users may attempt quizzes with multiple questions and different response times. Therefore, it is important that the system operates efficiently without errors and generates accurate results.

2.6.1 5.1 Constraints Identified

During the design and testing of the system, the following constraints were considered:

- Number of quiz questions and application performance
 - Accuracy of score calculation
 - Timer synchronization and response handling
 - User input validation and error handling
 - GUI responsiveness and usability
-

2.6.2 5.2 Handling of Constraints

To handle these constraints, the following solutions were implemented:

- Used efficient Python programming techniques for question management
- Implemented automatic score calculation and percentage generation
- Added timer functionality for quiz questions
- Applied input validation to handle incorrect user actions
- Performed debugging and testing to ensure smooth GUI operation
- Implemented error handling to prevent application crashes
- be simple and responsive

2.7 Handling of Constraints

To handle these constraints, the following solutions were implemented:

- Used efficient file handling using Python `os` and `shutil` modules
- Implemented a **progress bar** to show processing status
- Added **file counter** to track number of files processed
- Added checks to avoid overwriting duplicate files
- Ensured proper error handling for invalid folder selection

2.7.1 5.3 Test Plan / Test Cases

Test Case	Description	Expected Result
TC1	User Login	Login successful and quiz starts
TC2	Load Questions	Questions displayed correctly
TC3	Select Correct Answer	Score increases accordingly
TC4	Select Wrong Answer	Score remains unchanged
TC5	Timer Expiry	Next question loads automatically
TC6	Quiz Completion	Final score and percentage displayed correctly
TC7	Leaderboard Access	Stored scores displayed successfully

2.7.2 5.4 Test Procedure

The testing process was carried out as follows:

1. Launched the Quiz Game application.
2. Entered user details and started the quiz.
3. Answered multiple-choice questions using different inputs.
4. Verified score calculation for correct and incorrect answers.
5. Tested timer functionality and automatic question switching.
6. Completed the quiz and checked result generation.
7. Verified leaderboard and score storage functionality.

Multiple tests were performed to ensure consistent performance and accurate results.

2.7.3 5.5 Performance Outcome

The Quiz Game System performed successfully during testing.

- Questions were displayed correctly throughout the quiz.
 - Score calculation and percentage generation were accurate.
 - Timer functionality worked properly and responded as expected.
 - The graphical user interface remained responsive during operation.
 - User inputs were validated successfully.
 - No major errors or application crashes were observed during testing.
-

2.7.4 5.6 Limitations & Recommendations

Although the system works effectively, some limitations are:

- Limited number of predefined quiz questions.
- Supports only single-user operation.
- Basic leaderboard functionality.
- No online multiplayer support.

2.7.4.1 5.6.1 Future Improvements

- Add a larger database-driven question bank.
- Implement category-wise quizzes (Science, Sports, Programming, etc.).
- Add user authentication and profile management.

- Develop an online multiplayer quiz feature.
- Improve the user interface with additional themes and customization options.
- Integrate cloud-based score storage and performance analytics.